

On the inner-outer dichotomy of A-not-A questions:

A minimalist-cartographic account

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In this article, a fine-grained cartographic analysis of Mandarin A-not-A questions has been proposed to account for their interactions with various functional elements along the clausal spine: Inner A-not-A questions are sensitive to certain locality effects in Minimalist terms, whereas outer A-not-A questions are immune to the interference. It is proposed that the former combines two operations: The first step involves head movement to the head of PolP in overt syntax, which can be blocked by an intervening functional head. The second step involves LF feature movement to the Int head, which can be blocked by an intervening quantifier. By contrast, the latter resides in the left periphery, hence no modals or quantifiers are high enough to trigger blocking effects.

1. A-not-A in a nutshell

Chinese A-not-A questions have a lot of interesting properties with respect to their syntax, semantics and pragmatics, and are quite unique in typological terms. Following are the four properties of A-not-A question construals that stand out in the generative literature:

- I. They take a *wh*-question particle *ne*, instead of a yes-no question marker *ma*, as a sentence-final particle. In other words, they pattern with *wh*-questions rather than yes-no questions (cf. Huang 1982, Tang 1981, 1984, among others).
- II. They pattern with *wh*-adjuncts in displaying all sorts of locality effects, including those induced by strong islands, negative islands, and focus interventions (cf. Huang 1982, Aoun and Li 1993, Tsai 1994, Hagstrom 2006, among others).
- III. A-not-A questions may develop from alternative questions historically, but they differ crucially with respect to certain locality and ellipsis constraints. As a result, they cannot be treated alike (cf. Huang 1991).
- IV. Ernst (1994) and P. Law (2006) further point out that A-not-A questions are blocked by certain classes of adverbials, which presents a challenge for syntactic and semantic treatments alike (see also A. Law 2001, Jin 2016).

This paper presents an in-depth look at the last property (as well as the first three properties when necessary). We entertain a cartographic analysis to tackle the locality issues by adopting the feature-based Relativized Minimality advocated by Rizzi (2004), where the intervening factors have been reclassified into the following four types:

- I. Argumental: person, number, gender, case
- II. Quantificational: *Wh*, Neg, measure, focus...
- III. Modifier: evaluative, epistemic, Neg, frequentative, measure, manner, ...
- IV Topic

The first step, as specified in Section 2, is to distinguish an inner A-not-A question such as (1a) from an outer A-not-A question in (1b) in terms of syntactic cartography (cf. Rizzi 1997, Cinque 1999, Cinque and Rizzi 2010):

- (1) a. Akiu **qu-bu-qu** Rineiwa? (inner A-not-A)
 Akiu go-not-go Geneva
 ‘Will Akiu go to Geneva?’
- b. Akiu **shi-bu-shi** yao qu Rineiwa? (outer A-not-A)
 Akiu be-not-be will go Geneva
 ‘Is it true that Akiu will go to Geneva?’

Section 3 gives a comprehensive review of locality issues involved in construing A-not-A questions, in particular, those intervention effects discussed in Ernst (1994) and Law (2006). As illustrated in (2), it is proposed that outer A-not-A serves as the head of Assertion Phrase (AstP), sandwiched between IntP and EviP in the left periphery. By contrast, inner A-not-A merges to the *v*P periphery as the head of Polarity Phrase (PolP):

- (2) [IntP Q [AstP **A-not-A**^{outer} [EviP...[FinP...[PolP **A-not-A**^{inner} [*v*P ...]]]]]]

In Section 4, we take A-not-A questions to be a combination of two operations: the first part is head movement to check a [*u*V] feature in overt syntax, and the second part involves feature movement to check an [*i*Q] feature in LF. Section 5 explores relevant issues further along the lines laid out above. Section 6 concludes our arguments and analyses.

2. The inner-outer dichotomy of A-not-A questions

First note that inner A-not-A can be formed by a deontic modal like *yinggai* ‘should’, as in (3a), or a dynamic modal like *gan* ‘dare’, as in (3b). Interestingly, they themselves prevent the main verb *shuo* ‘speak’ from taking the A-not-A form, as evidenced by the ungrammaticality of (3c,d):

- (3) a. Akiu **ying-bu-yinggai**^{Deo} shuo yingwen? (deontic modal)
 Akiu should-not-should speak English
 ‘Should Akiu speak English?’
- b. Akiu **gan-bu-gan** shuo yingwen? (dynamic modal)
 Akiu dare-not-dare speak English
 ‘Does Akiu dare to speak English?’
- c. *Akiu **yinggai**^{Deo} shuo-bu-shuo yingwen?
 Akiu should speak-not-speak English
 ‘Should Akiu speak English?’
 [*deontic modal > inner A-not-A]
- d. *Akiu **gan** shuo-bu-shuo yingwen?
 Akiu dare speak-not-speak English
 ‘Does Akiu dare to speak English?’
 [*dynamic modal > inner A-not-A]

Outer A-not-A, on the other hand, may precede all types of modals without inducing any blocking effects, as exemplified by (4a-d):

- (4) a. Akiu shi-bu-shi **yinggai**^{Deo} shuo yingwen?
 Akiu be-not-be should speak English
 ‘Is it true that Akiu should speak English?’
 [outer A-not-A > deontic modal]
- b. Akiu shi-bu-shi **gan** shuo yingwen?
 Akiu be-not-be dare speak English
 ‘Is it true that Akiu dare speak English?’
 [outer A-not-A > dynamic modal]
- c. Akiu shi-bu-shi **yinggai**^{Epi} qu-le Rineiwa?
 Akiu be-not-be must go-Prf Geneva
 ‘Is it true that Akiu must have gone to Geneva?’
 [outer A-not-A > epistemic modal]
- d. Akiu shi-bu-shi **keneng** qu-le Rineiwa?
 Akiu be-not-be can go-Prf Geneva
 ‘Is it true that Akiu could have gone to Geneva?’

[outer A-not-A > epistemic modal]

By contrast, when *shi* ‘be’ is not present, only a verbal epistemic modal such as *keneng* ‘can’ may trigger the A-not-A morphology, as in (5a). In contrast to deontic *yinggai* ‘should’ in (3a), the same construal is unavailable for epistemic *yinggai*, as in (5b):¹

- (5) a. Akiu **ke-bu-keneng** qu-le Rineiwa? (verbal epistemics)
 Akiu may-not-may go-Prf Geneva
 ‘Is it true that Akiu may have gone to Geneva?’
 b. * Akiu **ying-bu-yinggai**^{Epi} qu-le Rineiwa? (non-verbal epistemics)
 Akiu must-not-must go-Prf Geneva
 ‘Must Akiu have gone to Geneva?’

(6a) shows that a speech-act adverb such as *laoshishuo* ‘honestly’ can precede inner A-not-A, whereas evidential adverbs like *jushuo* ‘allegedly’ and *haoxiang* ‘seemingly’ cannot, as evidenced by the deviance of (6b,c).

- (6) a. **laoshishuo** Akiu qu-mei-qu Rineiwa?
 honestly Akiu go-have.not-go Geneva
 ‘Honestly, did Akiu go to Geneva?’
 [speech act adverb > inner A-not-A]
 b. * Akiu **jushuo** qu-mei-qu Rineiwa?
 Akiu allegedly go-have.not-go Geneva
 ‘Did Akiu allegedly go to Geneva?’
 [*evidential adverb > inner A-not-A]
 c. * Akiu **haoxiang** qu-mei-qu Rineiwa?
 Akiu seemingly go-have.not-go Geneva
 ‘Did Akiu seemingly go to Geneva?’
 [*epistemic adverb > inner A-not-A]

¹ An epistemic modal is verbal if it can serve as a main predicate, as evidenced by the contrast between (ia) and (ib):

- (i) a. zhe-jian shi bu keneng.
 this-CL matter not possible
 ‘This matter is not possible.’
 b. * zhe-jian shi bu yinggai^{Epi}.
 this-CL matter not must.be.the.case
 ‘This matter must be the case.’
 c. zhe-jian shi bu yinggai^{Deo}.
 this-CL matter not allowed
 ‘This matter should not be allowed.’

It is also instructive to note that deontic *yinggai* in (3a) indeed has a predicate counterpart, as evidenced by (ic).

(7a) shows that outer A-not-A can be sandwiched between the speech-act adverb *laoshishuo* ‘honestly’ and the evidential adverb *jushuo* ‘allegedly’. (7b) shows a similar distribution with *haoxiang* ‘seemingly’.

- (7) a. **laoshishuo** Akiu shi-bu-shi **jushuo** qu-le Rineiwa?
 honestly Akiu be-not-be allegedly go-Prf Geneva
 ‘Honestly, is it true that Akiu allegedly went to Geneva?’
 [speech act adverb > outer A-not-A > evidential adverb]
- b. **laoshishuo** Akiu shi-bu-shi **haoxiang** qu-le Rineiwa?
 honestly Akiu be-not-be seemingly go-Prf Geneva
 ‘Honestly, is it true that Akiu seemingly went to Geneva?’
 [speech act adverb > outer A-not-A > epistemic adverb]
- c. * Akiu **jushuo** shi-bu-shi yao qu Rineiwa?
 Akiu allegedly be-not-be will go Geneva
 ‘Is it true that Akiu allegedly went to Geneva?’
 [*evidential adverb > outer A-not-A]
- d. * Akiu **haoxiang** shi-bu-shi yao qu Rineiwa?
 Akiu seemingly be-not-be will go Geneva
 ‘Is it true that Akiu seemingly went to Geneva?’
 [*epistemic adverb > outer A-not-A]

Likewise, merging a manner/frequency adverb to the left of inner A-not-A results in the ungrammaticality of (8a,b) (see Ernst 1994, Law 2006, Tseng 2009, among others):²

- (8) a. * Akiu **renzhende** nian-bu-nian shu?
 Akiu seriously study-not-study book
 ‘Does Akiu study hard?’
 [manner adverb > inner A-not-A]
- b. * Akiu **chang** qu-bu-qu Rineiwa?
 Akiu often go-not-go Geneva
 ‘Does Akiu often go to Geneva?’
 [frequency adverb > inner A-not-A]

Outer A-not-A, as expected, is immune to the blocking effects of *renzhende* ‘seriously’ and *chang* ‘often’, since it situates well beyond the scope of such adverbs, as evidenced by (9):

² Here *chang* ‘often’ can also be analyzed as a habitual aspect in semantic terms.

- (9) a. Akiu shi-bu-shi **renzhende** nianshu?
 Akiu be-not-be seriously study
 ‘Is it true that Akiu studies hard?’
 [outer A-not-A > manner adverb]
- b. Akiu shi-bu-shi **chang** qu Rineiwa?
 Akiu be-not-be often go Geneva
 ‘Is it true that Akiu often goes to Geneva?’
 [outer A-not-A > frequency adverb]

In sum, there are two factors involved in licensing inner and outer A-not-A construals: One concerns the hierarchical arrangement of various functional categories along the clausal spine, as sketched below:

- (10) *First Approximation* (> : precede and scope over)
 speech act adverb > *outer A-not-A* > evidential/epistemic modal > *inner A-not-A*
 > deontic modal > manner/frequency adverb > dynamic modal > main verb

Based on the above “topography” of A-not-A questions, it is not difficult to see why deontic and dynamic modals block V-not-V in (3c,d): This is because they are closer to inner A-not-A than the main verb, hence taking the priority of assuming the disjunctive morphology.

The second factor has to do with the intervention effects induced by certain classes of adverbs. For instance, V-not-V is blocked by the evidential/epistemic adverb in (6b,c), and by the manner/frequency adverb in (8a,b). We will come back to address these two issues in Section 4.

3. Previous analyses of blocking effects

In the literature, the most influential analysis of A-not-A questions (in particular, A-not-AB type) comes from Huang (1991): They are licensed by an implicit Q (or disjunctive) morpheme at the Infl head, which in turn triggers the hallmark reduplication morphology with a verbal element in its immediate scope, forming “A-not-A” on the surface. Schaffar and Chen (2001) makes a further contribution to the issue by teasing apart two types of A-not-A questions in terms of their distinct licensing contexts and syntactic distributions. For the outer A-not-A questions (C-neg-C in their terms), what is asked is the truth value of the proposition given the context below:

- (11) Context 1: I invite a friend to a restaurant called “Tabula Rasa” in Cologne, which I

have recently discovered. When we enter the restaurant, the waiter smiles at him. I ask my friend...

Ni shi-bu-shi lai-guo Tabula Rasa?

You be-not-be come-Exp Tabula Rasa

‘Have you been to the “Tabula Rasa” before?’

By contrast, the inner A-not-A question (V-neg-V in their terms) can be put forth ‘out of blue’, and construed as a disjunctive question:

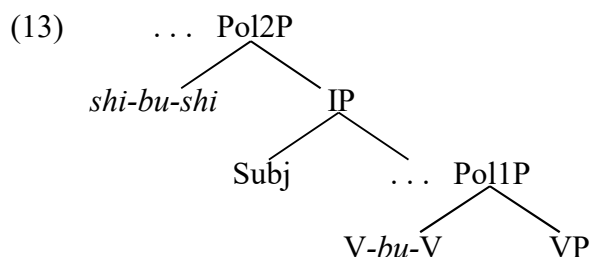
- (12) Context 2: I want to spend an evening together with a friend. After the cinema we are hungry. On our way home we pass by the Tabula Rasa. I ask him...

Ni lai-me-lai-guo Tabula Rasa?

You come-have-not-come-Exp Tabula Rasa

‘Have you been to the “Tabula Rasa” already?’

Their structural difference is illustrated in the following representation and its corresponding tree diagram: Outer A-not-A occupies the Spec of Pol2P, whereas inner A-not-A occupies the Spec of Pol1P, which is also the locus for verbal negation and narrow focus operators:



To account for the intervention effects observed in Section 2, Ernst (1994) suggests that the relative positioning of adverbs and the A-not-A operator is subject to the Isomorphic Principle given below (cf. Huang 1982, among others).

- (14) *The Isomorphic Principle:*

If an operator A has scope over B at S-Structure, then A has scope over B at LF.

As pointed out by Ernst, *yiding* ‘definitely’ is not compatible with inner A-not-A in (15a), the reason being that, after the Q morpheme undergoes the V-to-I-to-C movement at LF, *yiding* must QR to the top of the sentence to maintain the scope isomorphism, as in (15b):

- (15) a. * Ta yiding qu-bu-qu?

He definitely go-not-go

‘Is he definitely going?’

- b. LF: * *yiding_j* [*qu*+INFL_[+Qu]]_i [*ta t_j t_i*].

This in turn results in a semantic anomaly since a question cannot be definite, just like its English counterpart in (16):

- (16) * Definitely, is he going?

Law (2006), on the other hand, proposes that the A-not-A operator merges to VP first, and moves to the CP domain subsequently. An adverb will be the potential binder if it is semantically related to the predicate or its argument, or to the proposition in the scope of A-not-A operator. By contrast, temporal and locative adverbs are not potential binder, since they are related to the worlds and times instead of the proposition involved.

Take (17a) ((8a) repeated here)) for example: *renzhende* ‘seriously’ is related to the main predicate, and therefore counts as an intervener between *nian-bu-nian* ‘study-not-study’ and its trace at LF, as illustrated by (17b):

- (17) a. * Akiu **renzhende** nian-bu-nian shu?
 Akiu seriously study-not-study book
 ‘Does Akiu study hard?’
 b. LF: [_{CP} [*nian-bu-nian*]_k ... **renzhende** ... [_{VP} ... t_k ...]]
- 

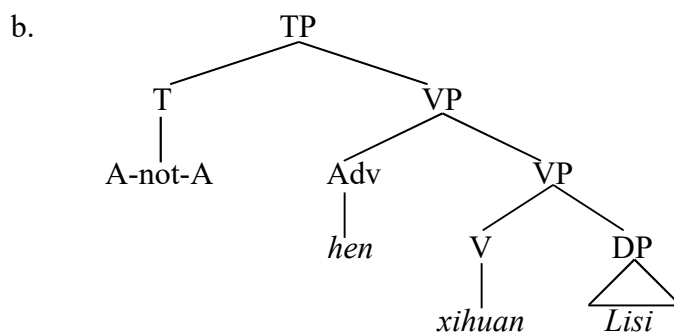
As a result, the sentence is ruled out by the Minimal Binding Requirement (18) along the line of Aoun and Li (1993):

(18) *The Minimal Binding Requirement (MBR)*

Variables must be bound by the most local potential antecedent (A’-binder).

From the perspective of PF, Tseng (2009) proposes that the A-not-A operator lowers from T to the morphosyntactic word that is immediately c-commanded by it (cf. Embick and Noyer 2001). Crucially, no head can intervene between A-not-A operator and its target, as illustrated in (19b). This accounts for the ill-formedness of (19a):

- (19) a. * Zhangsan hen xihuan-bu-xihuan Akiu?
 Zhangsan very like-not-like Akiu
 ‘Does Zhangsan like Akiu or not?’



Finally, R. Huang (2012) argues that the ungrammaticality of (20a) is due to an intervention effect caused by raising the A-not-A operator across the subject-oriented existential modal *you* ‘have’ at LF, as sketched in (20b):

- (20) a. * **you** yi-ge ren xi-bu-xihuan Akiu?
 have one-CL person like-not-like Akiu
 ‘Does a person like Akiu?’
- b. LF: [_{CP} [A-not-A]_k [_{IP} **you** Subj [_{VP} ... t_k ...]]]
-

If the A-not-A operator merges directly to the left periphery, assuming the form *shi-bu-shi* ‘be-not-be’, as in (21), then no existential operator is crossed, and no intervention effect is detected, as correctly predicted:

- (21) shi-bu-shi **you** yi-ge ren xihuan Akiu?
 be-not-be have one-CL person like Akiu
 ‘Is it the case or not that a person like Akiu?’

4. A cartographic account of the locality effects

It becomes clear at this stage that syntactic positions are as important as semantic/pragmatic considerations in determining the licensing conditions of A-not-A construals. Under the cartographic approach advocated by Rizzi (1997) and Cinque (1999), we propose to associate two types of A-not-A questions with two distinct interrogative projections according to “their height of interpretation”. Namely, we place outer A-not-A as the head of the Assertion Phrase (AstP) in the left periphery, and inner A-not-A in the Polarity Phrase (PolP) in the *v*P periphery (see also Belletti 2004, 2005; Tsai 2015), as illustrated below:

Consequently, the restriction of construing modals with inner A-not-A in (3a-d) is explained away: a modal like deontic *yinggai* ‘should’ undergoes overt head movement to PolP to check the [*uV*] feature, triggering the inner A-not-A morphology, as shown below:

$$(25) \dots [\text{PolP } ying\text{-}bu\text{-}yinggai^{Deo}_{[\#V]} [\text{MP } \langle yingai^{Deo} \rangle [\text{VP } \dots V \dots]]]$$

On the other hand, since a deontic/dynamic modal stands higher than the main verb, it blocks V-not-V in (3c,d) by virtue of being closer to the inner A-not-A head of PolP.

Moreover, since outer A-not-A is higher up in the clausal spine, it is by and large compatible with all sorts of modals in (4a-d), as illustrated below:

$$(26) \dots [\text{IntP } [iQ]\text{-Int}_{[\#Q]} \dots [\text{AstP } shi\text{-}bu\text{-}shi_{[\langle iQ \rangle, \#V]} [\text{EpiP } yinggai^{Epi} [\text{FinP } \dots]]]]$$

Interestingly, when *shi* is not present to occupy the seat of outer A-not-A, it is possible for a verbal epistemic modal to take its place, as in (27a) ((5a) repeated here): Being verbal, *keneng* ‘possible’ raises to Ast, triggering the outer A-not-A morphology, as illustrated in (27b):

(27) a. Akiu **ke-bu-keneng** qu-le Rineiwa? (outer A-not-A)
 Akiu may-not-may go-Prf Geneva

b. $\dots [\text{IntP } [iQ]\text{-Int}_{[\#Q]} \dots [\text{AstP } ke\text{-}bu\text{-}keneng_{[\langle iQ \rangle, \#V]} [\text{EpiP } \langle keneng \rangle [\text{FinP } \dots]]]]$

The same construal is unavailable for a non-verbal adverb such as epistemic *yinggai*, as we have seen in (5b).

As noted from the beginning, inner A-not-A questions are incompatible with certain types of adverbials, while their outer counterparts are immune from the blocking effects. To account for this curious restriction on inner A-not-A, we propose that the locus of a manner adverb like *renzhende* ‘seriously’ is in the Spec of ModP (modifier phrase, cf. Rizzi 2004), and that it is the implicit head Mod that blocks overt verb raising to inner A-not-A in (28a) ((8a) repeated here), as illustrated in (28b):

(28) a. *Akiu **renzhende** nian-bu-nian shu? (inner A-not-A)
 Akiu seriously study-not-study book

b. $[\text{FinP } Akiu [\text{PolP } nian\text{-}bu\text{-}nian_{[\langle iQ \rangle, \#V]} [\text{ModP } renzhende \textbf{Mod} [\text{VP } \langle nian \rangle \dots]]]]$

There are several ways to account for the locality displayed by V-to-Pol raising here. It can be attributed to the Head Movement Constraint (Travis 1984), the projection-based Relativized Minimality (Rizzi 1990), or Chomsky’s (1995) Attract Closest Condition. All in all, the

induced by *you* ‘have’, a subject-oriented existential modal, as we have seen in (20a,b).

It is also instructive to note that temporal/locative adverbials stand higher than PolP and do not count as quantificational. As a result, *mingtian* ‘tomorrow’ in (32a) blocks neither verb raising nor covert feature movement, as illustrated in (32b):

- (32) a. Akiu **mingtian** qu-bu-qu Rineiwa? (inner A-not-A)
 Akiu tomorrow go-not-go Geneva
- b. [IntP [*i*Q]-Int_[#Q] ... [FinP *mingtian* [FinP ... [PolP *qu-bu-qu*_{[<i>iQ>, #V</i>] [VP <*qu*> ...]]]]]]}
-

As for the speech act adverbial *laoshishuo* ‘honestly’ in (33a) ((6a) repeated here), it does not block movement of either type (i.e., overt head movement and/or covert feature movement). As illustrated in (33b), this is because it is hosted by the speech act phrase (SAP), and therefore situated above IntP (cf. the diagram (22)).

- (33) a. **laoshishuo** Akiu qu-mei-qu Rineiwa?
 honestly Akiu go-have.not-go Geneva
- b. [ASP **laoshishuo** [TopP Akiu [IntP [*i*Q]-Int_[#Q] ... [PolP *qu-mei-qu*_{[<i>iQ>, #V</i>] [VP <*qu*> ...]]]]]]}
-

5. Independent evidence for the minimalist-cartographic analysis

One piece of evidence for our minimalist-cartographic treatment comes from the fact that certain verbal elements at the vP periphery may optionally undergo inner A-not-A construals: Take (34a) for instance, *ba* may serve as a light verb, which carries a [*u*V] feature and triggers the inner A-not-A morphology by raising to the Pol head. Or it may serve as the head of a PP adverbial, as in (34b), which patterns with the temporal adverbial *mingtian* ‘tomorrow’ in not blocking the inner A-not-A question below:³

- (34) a. Akiu daodi **ba-bu-ba** Xiaodi dang yi-hui-shi?
 Akiu at.all BA-not-BA Xiaodi take one-CL-thing
 ‘Does Akiu take Xiaodi seriously at all?’

³ For people who find (34b) marginal, *ba* may indeed still keep its sore light verb status. On the other hand, for those who accept (34b), the following focus construal may also be acceptable: Here the *ba*-PP is fronted to the sentence-initial position, which should be impossible if *ba* can only serve as a light verb:

- (i) ? [PP *ba* Xiaodi], ni daodi dang-bu-dang yi-hui-shi?!
 BA Xiaodi you at.all take-not-take one-CL-thing
 ‘Do you take Xiaodi seriously at all?’

- b. Akiu daodi [PP **ba** Xiaodi] **dang-bu-dang** yi-hui-shi?
 Akiu at.all BA Xiaodi take-not-take one-CL-thing

This behavior is also attested with the following locative constructions: *zai* ‘at’ can be analyzed either as a light verb, triggering the inner A-not-A morphology, as in (35a), or as a preposition, which, unlike a deontic/dynamic modal or a manner/frequency adverb, does not block a V-not-V construal, as evidenced by (35b):

- (35) a. Akiu **zai-bu-zai** jia zuo fan?
 Akiu at-not-at home do meal
 ‘Does Akiu cook at home?’
 b. Akiu [PP **zai** jia] **zuo-bu-zuo** fan?
 Akiu at home do-not-do meal

On the other hand, if a verbal element is unambiguously treated as a light verb, then it must assume the A-not-A form itself, as is the case with *rang* ‘let’ in (36a). Otherwise, it would block the inner A-not-A construal of the main verb, thus crashing the derivation of (36b):

- (36) a. Laoshi **rang-bu-rang** Akiu qu Rineiwa?
 Teacher let-not-let Akiu go Geneva
 ‘Did the teacher let Akiu go to Geneva?’
 b. * Laoshi **rang** Akiu **qu-bu-qu** Rineiwa?
 Teacher let Akiu go-not-go Geneva

The same pattern is attested for a semi-functional category like *yong* ‘use/with’, as shown by the contrast between (37a,b):

- (37) a. Akiu **yong-bu-yong** daozi qie rou?
 Akiu with-not-with knife cut meat
 ‘Does Akiu cut meat with knife?’
 b. * Akiu yong daozi **qie-bu-qie** rou?
 Akiu with knife cut-not-cut meat

Interestingly enough, when *renzhen-de* ‘seriously’ loses its adverbial marker (AM) *-de*, the inner A-not-A construal becomes a manner predicate, hence subject to the inner A-not-A construal of (38a). The same usage, as expected, is blocked with the original adverbial form, as in (38b):

- (38) a. Akiu pingchang **ren-bu-renzhen** nian shu?
 Akiu usually serious-not-serious study book
 ‘Does Akiu study hard usually?’
- b. * Akiu pingchang **ren-bu-renzhen-de** nian shu?
 Akiu usually serious-not-serious-AM study book
 ‘Does Akiu study hard usually?’

Another argument has to do with a higher version of the intervention effects in the left periphery. As mentioned above, the verbal epistemic modal *keneng* ‘possible’ in (39a) is subject to the outer A-not-A usage. This construal, however, is blocked by an evidential adverb such as *xianran* ‘obviously’ in (9). Again, the sentence can be rescued with *shi* merged directly to outer A-not-A at the Ast head, scoping over the evidential adverb *xianran*, as in (39c):

- (39) a. Akiu **ke-bu-keneng** hui qu Rineiwa?
 Akiu possible-not-possible will go Geneva
 ‘Is it (obviously) possible for Akiu to go to Geneva?’
- b. * Akiu **xianran** ke-bu-keneng hui qu Rineiwa?
 Akiu obviously possible-not-possible will go Geneva
 ‘Is it obviously possible for Akiu to go to Geneva?’
- c. Akiu **shi-bu-shi** xianran keneng hui qu Rineiwa?
 Akiu be-not-be obviously possible will go Geneva
 ‘Is it true that it is obviously possible for Akiu to go to Geneva?’

To visualize the derivations in question, (40a) represents the two steps of forming the outer A-not-A question of (39a) in the left periphery. Along this line, (39b) is ruled out simply because it is impossible for *keneng* ‘possibly’ to cross over the evidential operator *xianran* ‘obviously’, as illustrated in (40b). On the other hand, given that *shi* is merged directly to Ast, there is nothing to block the subsequent movement of [*iQ*] to Int, as in (40c). We therefore correctly predict that (39a,c) is grammatical while (39b) is not:

- (40) a. $[_{\text{IntP}} [iQ] - \text{Int}_{[\#Q]} \dots [_{\text{AstP}} M\text{-not-}M_{[\langle iQ \rangle, \#V}] [_{\text{MP}} \langle M \rangle \dots]]]$
- b. $[_{\text{IntP}} [iQ] - \text{Int}_{[\#Q]} \dots [_{\text{AstP}} M\text{-not-}M_{[\langle iQ \rangle, \#V}] [_{\text{EviP}} \text{Adv} [_{\text{MP}} \langle M \rangle \dots]]]]$
- c. $[_{\text{IntP}} [iQ] - \text{Int}_{[\#Q]} \dots [_{\text{AstP}} \text{shi-not-shi}_{[\langle iQ \rangle, \#V}] [_{\text{EviP}} \text{Adv} [_{\text{MP}} M [\dots]]]]]]$

6. Further Issues

Given our inner-outer dichotomy of A-not-A questions in Chinese, one might raise the objection that sentences with outer A-not-A should be considered as bi-clausal (or even multi-clausal if all categories subject to A-not-A construals count as full-fledged lexical verbs). This line of thinking, however, misses an important generalization that outer A-not-A questions are speaker-oriented and infelicitous in an out-of-blue context (cf. Schaffar and Chen 2001), a sure indication of peripheral construals. We also need to heed the fact that quite a few “odd” elements may assume the A-not-A form: In addition to those semi-functional categories like modals and light verbs, *chang* ‘often’ can also be associated with inner A-not-A, presumably serving as a habitual aspect rather than a genuine adverb:

- (41) Akiu zuijin **chang-bu-chang** qu Rineiwa?
 Akiu lately often-not-often go Geneva
 ‘Does Akiu often go to Geneva lately?’

Under the minimalist-cartographic approach, we not only provide a syntactic account of the pragmatics of outer A-not-A questions, but also map out the distribution of a variety of (semi-)functional elements along the clausal spine.

Another point of interest has to do with the issue why only deontic *yinggai* ‘should’, but not its epistemic counterpart, is able to assume the A-not-A form. First note that deontic *yinggai* licenses VP ellipsis, as in (42), while epistemic *yinggai* does not, as in (43):

- (42) Akiu **yinggai**^{Deo} qu Rineiwa, Xiaodi ye **yinggai**^{Deo} ~~qu~~ ~~Rineiwa~~.
 Akiu should go Geneva Xiaodi too should go Geneva
 ‘Akiu should go to Geneva, and Xiaodi should, too.’
- (43) *Akiu **yinggai**^{Epi} qu-le Rineiwa, Xiaodi ye **yinggai**^{Epi} ~~qu-le~~ ~~Rineiwa~~.
 Akiu should go-Prf Geneva Xiaodi too should go-Prf Geneva
 ‘Akiu should have gone to Geneva (by now), and Xiaodi should have, too.’

This shows that the former is much more verbal than the latter as the head licenser of the ellipsis sites. The distinction receives further confirmation from the licensing capacity of *keneng* ‘may’, a verbal epistemic that triggers the outer A-not-A morphology in (5a) and (27a). As illustrated in (44), *keneng*, just like deontic *yinggai*, can license the ellipsis site below (presumably FinP or TP, with *Akiu* and *Xiaodi* as the contrastive topic):

- (44) Akiu **keneng** qu-le Rineiwa, Xiaodi ye **keneng** ~~qu-le~~ ~~Rineiwa~~.
 Akiu may go-Prf Geneva Xiaodi too may go-Prf Geneva
 ‘Akiu could have gone to Geneva, and Xiaodi could have, too.’

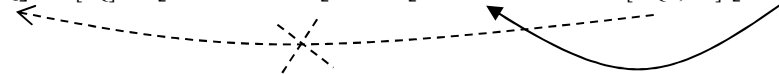
7. Concluding Remarks

To sum up, we have laid out a fine-grained cartographic analysis of the two types of A-not-A questions based on their interactions with modals, adverbials and light verbs with respect to their height of interpretation. This move accounts for the fact that only outer A-not-A is speaker-oriented and infelicitous in an out-of-blue context, whereas inner A-not-A is more sensitive to the blocking effects from all sorts of (semi-)functional elements.

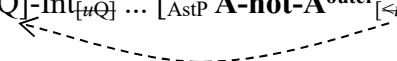
Crucially, we take an A-not-A question to be a combination of two operations: The first part involves head movement to check the $[uV]$ feature on the Pol head in overt syntax. More specifically, $[uV]$ is checked off by attracting the closest verbal head, which in turn assumes the A-not-A form. This step can be blocked by an intervening head like a modal or a modifier head, as shown below:

- (45) ... [_{PolP} **V-not-V**_[<iQ>, #V] ... [**Modal/Modifier** ... [_{VP} <V> ...]]]
- 

The second part involves feature movement to check the $[iQ]$ feature on the Int head in LF: $[iQ]$ raises further to Int to check off $[uQ]$ (cf. Chomsky 2000, among others), and can be blocked by an intervening quantifier. As sketched in the following diagram, the covert feature raising is ruled out in violation of the feature-based Relativized Minimality:

- (46) [_{IntP} [_{iQ}]-Int_[#Q] ... [**Quantifier** [_{FinP} ... [_{PolP} **A-not-A**^{inner}_[<iQ>, #V] [_{VP} ...]]]]]
- 

By contrast, outer A-not-A questions have *shi* merged high to the Ast head in the left periphery, as schematized below:

- (47) [_{IntP} [_{iQ}]-Int_[#Q] ... [_{AstP} **A-not-A**^{outer}_[<iQ>, #V] ... [_{FinP} ...]]]
- 

As a result, no modals or quantifiers are high enough to block the Ast-to-Int raising of $[iQ]$. This explains why outer A-not-A is immune to all the locality effects considered in our study.

Acknowledgement

This article is a tribute to Professor James Huang for his lifelong dedication to our cause. We are grateful to Adriana Belletti, Richard Kayne, Paul Law, Luigi Rizzi, Fuzhen Si and Chen Zhao for sharing their insights. Special thanks to Andrew Simpson and two anonymous reviewers for their helpful comments and suggestions. The research leading to this article is funded by the Ministry of Science and Technology of Taiwan (MOST 106-2410-H-007-030-MY3 and MOST 109-2410-H-007-065-MY2) and the National Science and Technology Council of Taiwan (NSTC 113-2410-H-007-043-MY3 and NSTC 113-2410-H-005 -021).

References

- Aoun, Joseph and Li, Yen-Hui Audrey. 1993. *Syntax of Scope*. Cambridge, MA: MIT Press.
- Belletti, Adriana. 2004. Aspects of the low IP area. In *The Structure of CP and IP*, Luigi Rizzi (eds), 16–51. Oxford: Oxford University Press.
- Belletti, Adriana. 2005. Extended doubling and the VP periphery. *Probus* 17: 1: 1–35.
- Chomsky, Noam. 1995. *The Minimalist Program*. Cambridge, MA: MIT Press.
- Chomsky, Noam. 2000. Minimalist inquiries: the framework. In *Step by Step: Essays on Minimalism in Honor of Howard Lasnik*, Roger Martin, David Michaels & Juan Uriagereka (eds.), 89–155. Cambridge, MA: MIT Press.
- Cinque, Guglielmo. 1999. *Adverbs and Functional Heads*. Oxford: Oxford University Press.
- Cinque, Guglielmo and Rizzi, Luigi. 2010. The cartography of syntactic structures. In *Oxford Handbook of Linguistic Analysis*, Bernd Heine & Heiko Narrog (eds.), 51–65. Oxford: Oxford University Press.
- Embick, David and Noyer, Rolf. 2001. Movement operations after syntax. *Linguistic Inquiry* 32: 4: 555–595.
- Ernst, Thomas. 1994. Conditions on Chinese A-not-A questions. *Journal of East Asian Linguistics* 3: 241–264.
- Hagstrom, Pau. 2006. A-not-A question. In *The Blackwell Companion to Syntax*, vol. 1, Martin Everaert and Henk van Riemsdijk (eds), 173–213. Oxford: Blackwell.
- Huang, C.-T. James. 1982. Move WH in a language without WH movement. *Linguistic Review* 1, 369–416.
- Huang, C.-T. James. 1991. Modularity and Chinese A-not-A questions. In *Interdisciplinary Approaches to Language: Essays in Honor of S.-Y. Kuroda*, Carol Georgopoulos and Roberta Ishihara (eds), 305–332. Dordrecht: Kluwer Academic Publishers.
- Huang, Rui-heng Ray. 2012. On two types of existential subjects in Chinese A-not-A questions. *Language and Linguistics* 13: 6: 1171–1210.

- Jin, Dawei. 2016. *The Semantics-Pragmatics Interface and Chinese Island Constraints*. PhD dissertation, The State University of New York at Buffalo.
- Law, Ann. 2001. A-not-A questions in Cantonese. *UCL working papers in linguistics* 13: 295–318.
- Law, Paul. 2006. Adverbs in A-not-A questions in Mandarin Chinese. *Journal of East Asian Linguistics* 15: 97–136.
- Pesetsky, David. 2000. *Phrasal Movement and its Kin*. Cambridge, MA: MIT Press.
- Rizzi, Luigi. 1990. *Relativized minimality*. MA: MIT Press.
- Rizzi, Luigi. 1997. The fine structure of the left periphery. In *Elements of Grammar*, Liliane Haegeman (eds), 281–338. Dordrecht: Kluwer.
- Rizzi, Luigi. 2004. Locality and the left periphery. In *Structures and Beyond. The Cartography of Syntactic Structures*, vol. 3, Adriana Belletti (eds), 223–251. New York: Oxford University Press.
- Schaffar, Wolfram and Chen, Lansun. 2001. Yes-No questions in Mandarin and the theory of focus. *Linguistics* 39: 837–870.
- Soh, Hooi Ling. 2005. *Wh*-in-situ in Mandarin Chinese. *Linguistic Inquiry* 36: 143–155.
- Tang, Ting-Chi. 1981. Guoyu yiwenju de yanjiu [The study of questions in Chinese]. *Shida Xubao* [Journal of Taiwan Normal University] 26: 219–277.
- Tang, Ting-Chi. 1984. Guoyu yiwenju yanjiu xulun [Question in Chinese revisited]. *Shida Xubao* [Journal of Taiwan Normal University] 29:381–435.
- Travis, Lisa. 1984. *Parameters and Effects of Word Order Variation*. PhD dissertation, MIT.
- Tsai, Wei-Tien Dylan. 1994. On nominal islands and LF extraction in Chinese. *Natural Language and Linguistic Theory* 12: 121–175.
- Tsai, Wei-Tien Dylan. 2015. On the topography of Chinese modals. In *Beyond Functional Sequence*, Uri Shlonsky (eds), 275–294. New York: Oxford University Press.
- Tseng, Wen-Hsin Karen. 2009. A Post-syntactic approach to the A-not-A questions. *USTWPL* 5: 107–139.